RAWild.

Toward Sensor-Agnostic RAW Object Detection
via Physics-Guided Curve and Grid Modeling

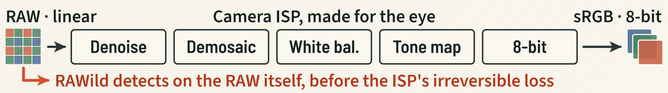
Shuhong Liu[1,2*] · **Gengjia Chang**[2*] · **Jun Liu**[2] · **Xuangeng Chu**[1,2] · **Yinqiang Zheng**[1] · **Tatsuya Harada**[1,3] · **Ziteng Cui**[1†]

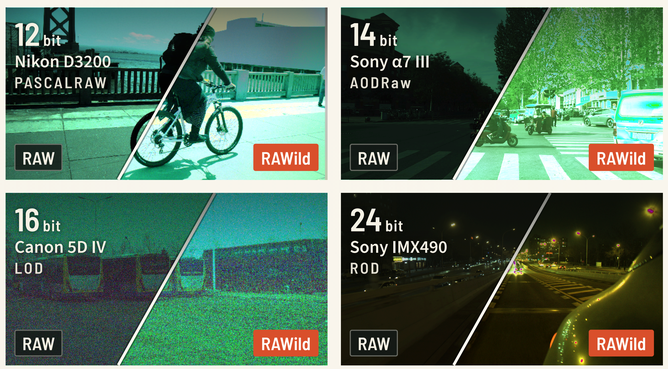
[1]The University of Tokyo [2]I2WM [3]RIKEN [*]Equal contribution [†]Corresponding author

NEURAL INFORMATION PROCESSING SYSTEMS 2026

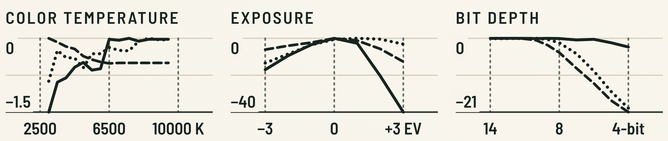
1 Motivation & Contribution

What is RAW? The sensor's linear 10–24-bit readout, before the ISP renders 8-bit sRGB for the eye. RAW keeps the detail lost in the dark and in glare, **but every sensor's RAW differs.**





One factor is enough to drop a linear-RAW detector:



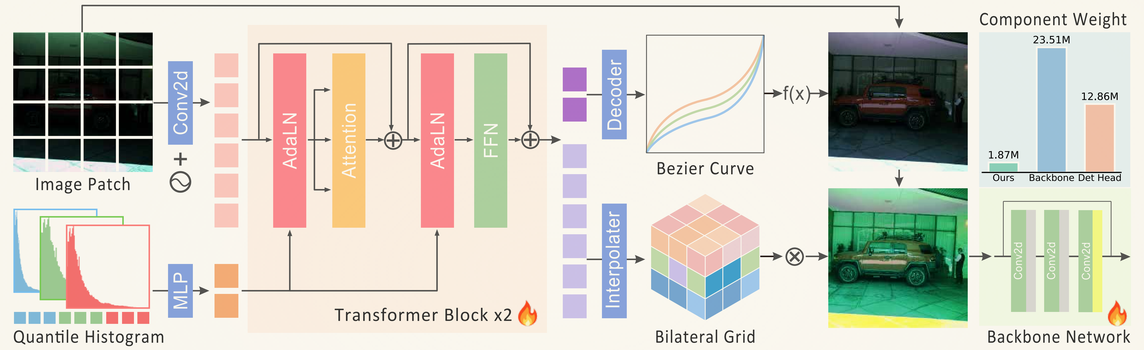
Learned ISPs fit one sensor and break when sensors mix.

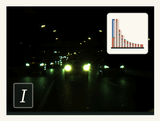
1. **RAWild**, a histogram-guided adapter that splits sensor variation into a **global Bézier curve** and a **local bilateral grid**.
2. A **physics-based RAW simulator** that renders labeled RAW as new virtual sensors.
3. **Best mAP@50 on all five mixed-sensor sets.**

2 Model Structure

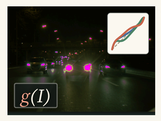
A global curve, then a local grid.

RAWild sits in front of a pretrained RetinaNet; both are fine-tuned for detection. A histogram-guided Transformer predicts a global Bézier curve g for tone (exposure, bit depth, color temperature) and a local bilateral grid M for color (mixed light, shading, crosstalk): $y(p) = M(p, \ell(p))\, g(I(p))$, with ℓ the luminance of I.





1 Histogram-guided. 64 quantiles per channel profile the sensor and modulate every block (AdaLN).



2 Δ-Bézier curve g. Ends pinned at 0 and 1; only a bounded offset δ from the identity is learned.



3 Grid $M = D(\mathbf{I} + A)$. Gain D times identity plus mixing $|A| \le .05$, so M stays invertible; read at (p, ℓ).

3 Physics-based RAW Synthesis

Simulate any sensor.

Real RAW covers only a few cameras. From any labeled RAW frame we render **virtual sensors**: sample a camera's spectra and an illuminant, fit a 3×3 matrix on a 24-patch color chart to swap them in, then add exposure, crosstalk, roll-off and quantization. Every step is per-pixel, so the boxes carry over.

Spectra 28 cams → PCA ➔ **Illuminant** 2500–20000 K ➔ **Exposure** ±3 EV ➔ **Crosstalk** ±.05 · black ±.02 ➔ **Roll-off** soft clip 0.9–1 ➔ **Quantize** 10–16 bit



4 Results

One network, every sensor.

Mixed sensors, one model per set

mAP@50 · ms (H100)

	SYNTHETIC			REAL · MIXED		SPEED
	PAS	LOD	ROD	PAS+LOD	Multi-RAW	ms ↓
Linear-RAW	89.21	53.10	34.97	60.57	25.39	**7.20**
RAW-Adapter	88.69	58.74	36.37	65.40	23.34	10.00
Dr.RAW	89.49	59.24	42.34	66.60	25.72	10.52
Dark-ISP	88.88	58.37	37.03	64.20	26.03	20.45
RAWild	**89.53**	**66.46**	**42.91**	**68.65**	**35.28**	10.70

Synthetic sets come from block 3. RAWild is best on all five.

Per sensor, trained separately

RetinaNet R-50 · mAP@50

	PASCALRAW · 12-BIT			16-BIT	24-BIT	14-BIT
	Low	Normal	Over	LOD	ROD	AODRaw
Linear-RAW	76.68	86.61	87.47	57.10	30.41	23.01
Default-ISP	83.90	89.59	89.67	58.80	41.78	32.41
RAW-Adapter	86.47	89.42	89.87	60.82	39.50	29.34
AdaptiveISP	88.71	90.14	90.11	61.42	43.79	33.82
Dark-ISP	71.39	89.40	88.40	55.60	40.90	32.40
Dr.RAW	88.65	89.83	89.82	64.08	47.20	**36.30**
RAWild	**89.11**	**90.19**	**90.20**	**69.09**	**55.41**	36.23

Best on 5/6 with ResNet-50 (AODRaw −0.07) and on 6/6 with Swin-T.

Ablation, one design at a time

LOD · R-50 · mAP@50

Histogram	None	Prefix	AdaLN
	68.38	63.97	69.09
Curve	None	Vanilla	Δ-Bézier
	66.50	65.11	69.09
Grid	None	Vanilla	D	I+A	D(I+A)
	66.91	66.73	65.89	66.46	69.09

Removing or swapping any design lowers mAP@50.

5 Detection Across Sensors

Fewer missed boxes in the dark and in glare.

